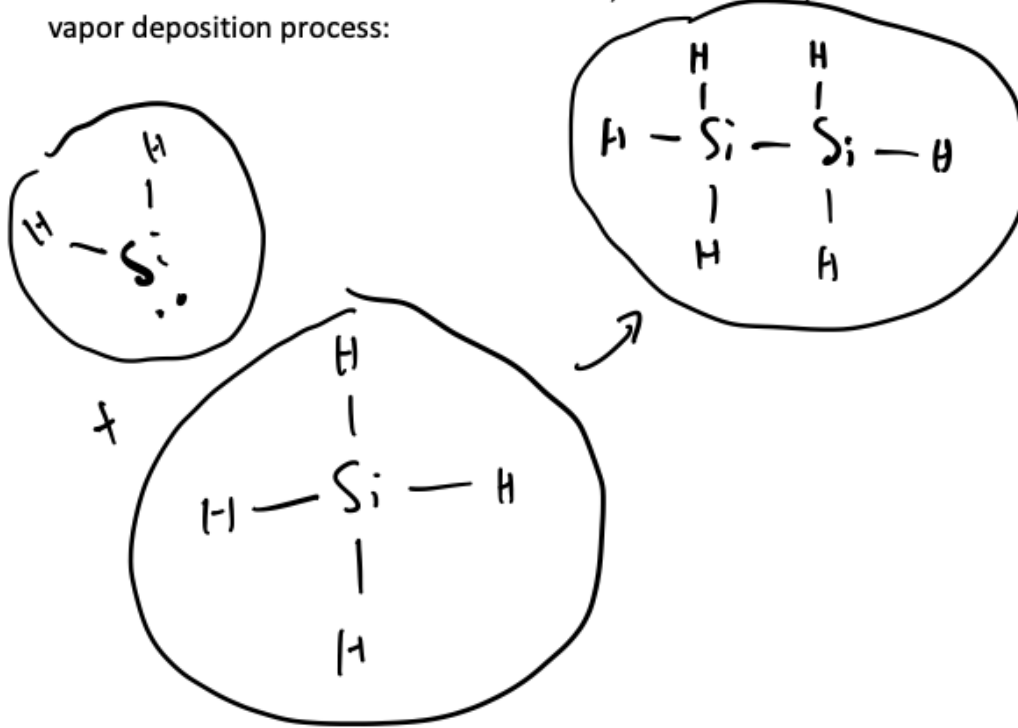


## 33 Reactions at the Molecular Level

Notes from <https://sites.engineering.ucsb.edu/~jbrow/chemreactfun/ch2/slides-stoi-2up.pdf>

Consider the reaction  $\text{SiH}_4 + \text{SiH}_2 \rightleftharpoons \text{Si}_2\text{H}_6$ , which is a step in a chemical vapor deposition process:



The reaction *extent*  $\xi$  keeps track of the number of times this reaction event occurs, i.e., it counts

the number of times an  $\text{SiH}_4$  molecule combines with an  $\text{SiH}_2$  molecule and forms a  $\text{Si}_2\text{H}_6$  molecule

The reaction *rate* is then

$$r = \frac{\Delta \xi}{\Delta t V} \quad \text{where } \Delta t \text{ is "small" } (10^{-13} - 10^{-12} \text{ s})$$

Where  $V$  is large enough to comprise multiple molecules but small enough that it can be

assumed to be well-mixed

If  $\Delta\xi$  is positive:

reaction proceeds in the forward direction

If  $\Delta\xi$  is negative:

reaction proceeds in the reverse direction

the reaction  
If ~~the~~ is in equilibrium:  $\Delta\xi = 0$ , equal number  
of forward events  $\frac{1}{2}$  reverse events

Note that for this reaction, it is easy to "link" the rate of reaction to the rate of generation or consumption of one of the reactant or product species, since

$$r = -r_{\text{SiH}_4} = -r_{\text{SiH}_2} = r_{\text{SiH}_6}$$

We can thus measure the reaction rate based on the change in species

concentrations, i.e.,  $\frac{1}{\nu} \frac{d\xi}{dt} \propto \frac{dC_j}{dt}$

However, for most reactions, it is not possible to measure the reaction rate. We hence must measure the rates of generation or production of individual species. This is why rates are often written

in terms of specific reactant or product species

An example of such a reaction is  $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ , where

$$r = -r_{\text{NO}} = -2r_{\text{O}_2} = r_{\text{NO}_2}$$

Consider the reaction  $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ . Is it likely that this reaction could occur as a molecular event? Why or why not?

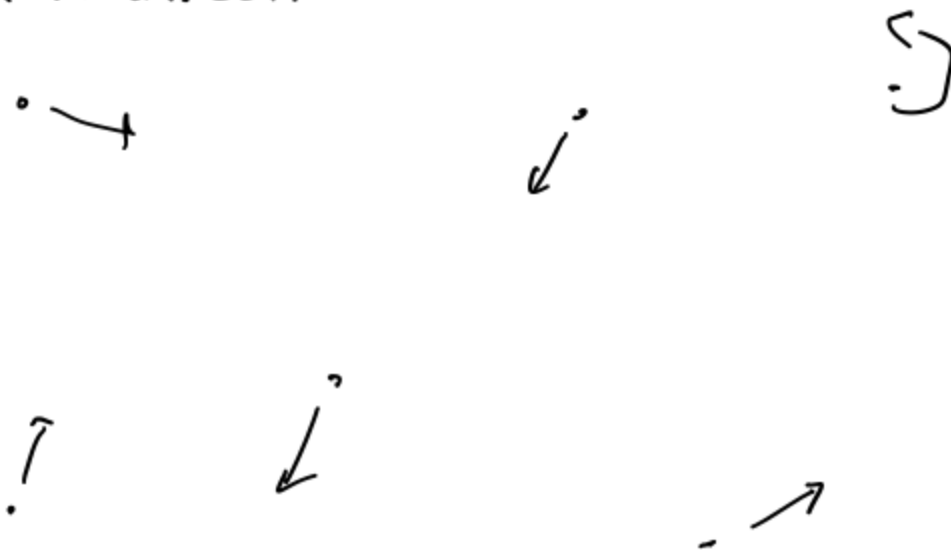
Let's answer this question by considering what must happen on the molecular level for the reaction to occur:

- $\text{O}=\text{O}$  bond must break

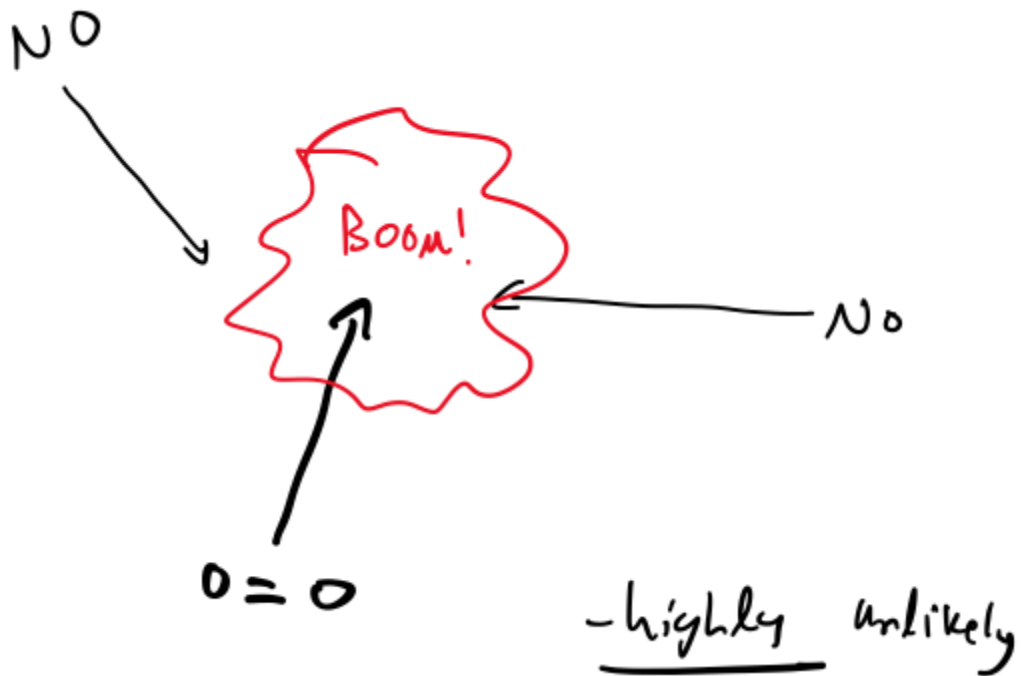
- two  $\text{O}-\text{NO}$  bonds must form

Consider a homogeneous gas phase reaction. The gas molecules:

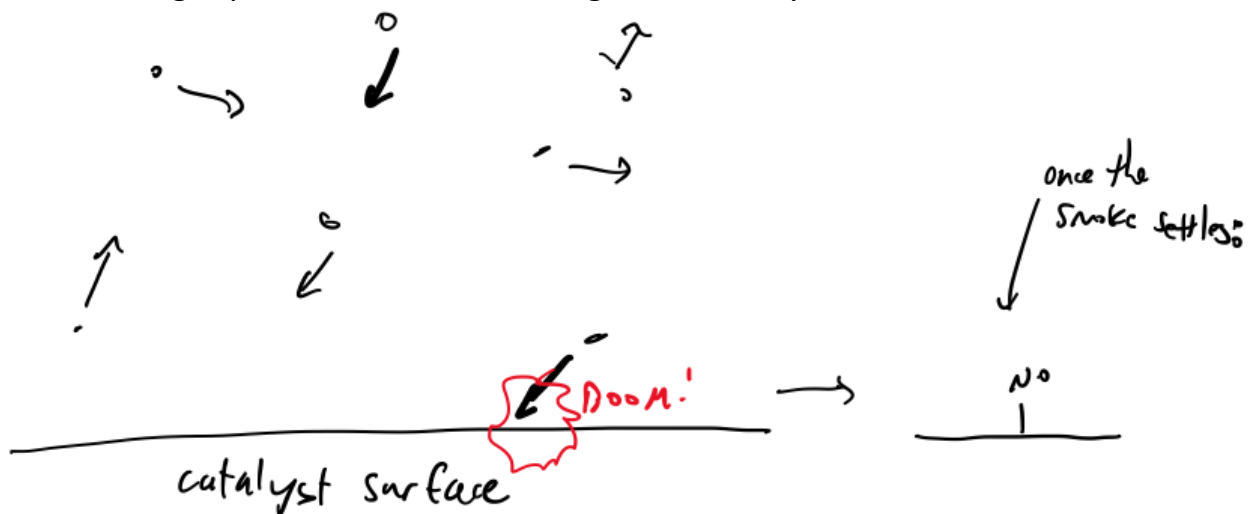
are moving at fast speeds in somewhat random directions.



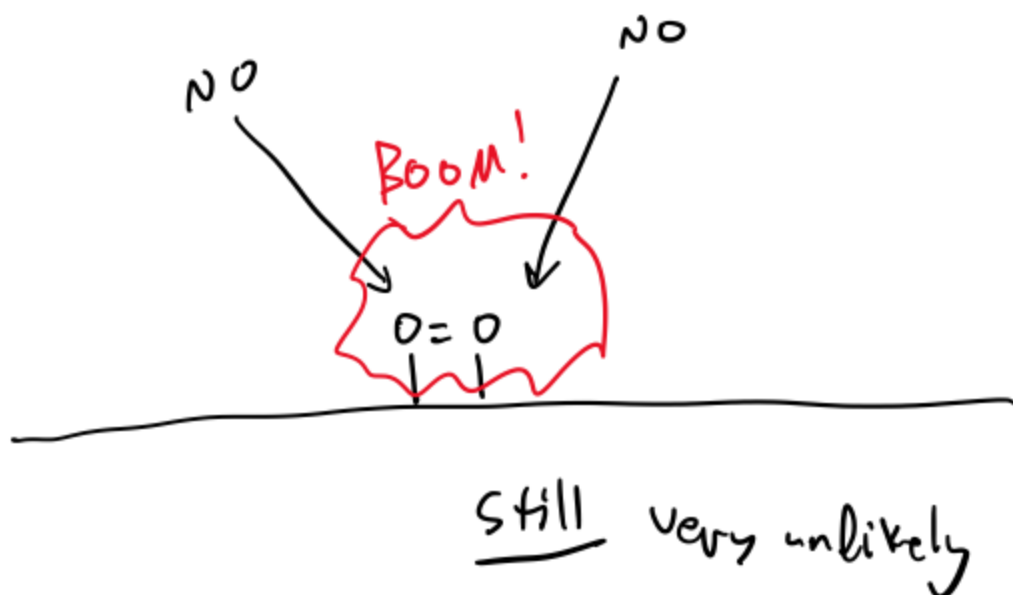
For the reaction  $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$  to happen as one molecular event:



Consider a gas phase reaction occurring over a catalyst surface:



For the reaction  $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$  to happen as one molecular event:



So, to answer our question, no, this reaction is not likely to happen as a molecular event because

the likelihood of 3 molecules colliding, let alone at an orientation that would favor reaction, is TINY.

So, we've looked at two reactions so far,  $\text{SiH}_4 + \text{SiH}_2 \rightleftharpoons \text{Si}_2\text{H}_6$  and  $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ . What if I told you that the reaction rates for these reactions were:  $r_{\text{Si}_2\text{H}_6} = k_{\text{Si}_2\text{H}_6} C_{\text{SiH}_4} C_{\text{SiH}_2}$  and  $r_{\text{NO}_2} = k_{\text{NO}_2} C_{\text{NO}}^2 C_{\text{O}_2}$  but that the first reaction is elementary but the second one is not 🤔

Reactions tend to convey two different meanings:

1) the change in the relative amounts of species that is observed when a reaction proceeds

2) The actual molecular events that happen as the reaction proceeds

The first is a stoichiometry statement the second is an elementary reaction



An elementary reaction is a change from reactants to products that proceeds without the formation of intermediate species

Elementary reactions are usually unimolecular or bimolecular because

the probability of species coming into contact decreases as the number of species increases

If the rate of a non-elementary reaction happens to exhibit elementary kinetics, it is

purely coincidental

I.e., the observed rate of  $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$  is  $r_{\text{NO}_2} = k_{\text{NO}_2} C_{\text{NO}}^2 C_{\text{O}_2}$ , this means that the

reaction  $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$  exhibits elementary kinetics however, the

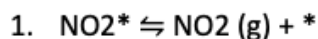
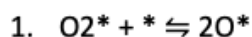
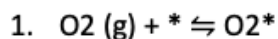
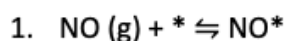
reaction  $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$  is not an elementary reaction since

it cannot proceed as a single molecular event

Rather, this reaction likely occurs via a reaction mechanism

which is comprised of multiple elementary steps

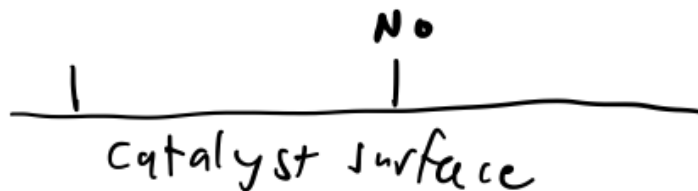
For example, consider the case where  $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$  occurs over a catalyst surface. One possible mechanism is:



$*$  = open catalyst site

$\text{NO}^*$  = NO bound to catalyst

$\text{O}_2^*$                        $\text{NO}_2^*$



the overall reaction  $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$  is a linear combination

of reactions 1-5. The overall stoichiometry is reproduced by adding

$$2 \times \text{rxn 1} + \text{rxn 2} + \text{rxn 3} + 2 \times \text{rxn 4} + 2 \times \text{rxn 5}.$$

Turning this into a vector:

$$s = \begin{bmatrix} \text{rxn 1} & \text{rxn 2} & \text{rxn 3} & \text{rxn 4} & \text{rxn 5} \\ 2 & 1 & 1 & 2 & 2 \end{bmatrix}$$

Writing the stoichiometry matrix for the reaction mechanism:

$$v = \begin{bmatrix} \text{NO} & \text{NO}^* & \text{O}_2 & \text{O}_2^* & \text{O}^* & \text{NO}_2 & \text{NO}_2^* & * \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1 & 2 & 0 & 0 & -1 \\ 0 & -1 & 0 & 0 & -1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 1 \end{bmatrix}$$

We can then multiply  $s$  by  $v$  to obtain

$$\begin{bmatrix} \text{NO} & \text{NO}^* & \text{O}_2 & \text{O}_2^* & \text{O}^* & \text{NO}_2 & \text{NO}_2^* & * \\ -2 & 0 & -1 & 0 & 0 & 2 & 0 & 0 \end{bmatrix}$$

which is the overall stoichiometry statement for the reaction, i.e.,

